LESSON PLAN

AIRCRAFT WIRING PRACTICES (ELECTRICAL WIRING AND INTERFACE SYSTEM) (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Main rotor movements, symmetry of lift feathering, flapping, dragging, hook's joint effect, flapping to equality, phase lag and advance angle
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Main rotor movements, symmetry of lift feathering, flapping, dragging, hook's joint effect, flapping to equality, phase lag and advance angle concepts.
5. Specific Objectives	12.1.5.1 Trainee will be able to understand Main rotor movements, symmetry of lift feathering, flapping, dragging, hook's joint effect, flapping to equality, phase lag and advance angle concepts.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of rotor dynamics diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary movements of a main rotor?
(b) Define symmetry of lift in rotor systems.
(c) List the factors causing flapping and dragging.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Main rotor movements overview • Symmetry of lift feathering • Flapping and dragging dynamics • Hook's joint effect analysis • Flapping to equality principle	L & Ds	PPT, IB	15 min
(b)	• Phase lag concept explanation • Advance angle mechanics • Rotor system equilibrium • Lift distribution analysis • Dynamic stability factors	L & Ds	PPT, PDF	20 min
(c)	• Integration of rotor movements • Application of phase lag theory • Summary of flapping dynamics • Review of hook's joint effect • Final concept reinforcement	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between phase lag and advance angle.
(b) Describe how flapping contributes to rotor symmetry.
(c) Identify the role of the hook's joint in rotor dynamics.

12. Home Assignment and Follow Up	Prepare a brief note on how rotor movements affect hovering and forward flight, linking concepts to LP 06 syllabus.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Hovering and forward flight, take off and climb to free air hover, vertical decent ground effect ground cushion re circulation effects.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min